

EBU6505 – Reasoning and Agents

Block 1, Day 4 – Knowledge Graphs in Modern AI

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Teaching Week 3

What have we covered so far this week?

- MDPs describe sequential decision problems.
- Sequential decision models use reward to compare future choices.
- Knowledge representation stores facts in a machine-usable form.
- Knowledge graphs connect those facts into useful paths.

Main idea: Day 3 showed how to build a graph. Today shows how modern AI systems use that graph.

What are we going to cover today?

- How knowledge graphs help retrieval, memory, and explanation
- How graphs fit inside modern AI systems and agent pipelines
- What neuro-symbolic AI means in simple terms
- Why MCP, multi-agent systems, and A2A matter in practice

Main idea: The main question today is not how to build a graph. It is how to use a graph well inside a real AI system.

- *Artificial Intelligence: Foundations of Computational Agents*, Poole and Mackworth (3rd ed.)
- Chapters 16 and 17 – graph structure and reasoning with linked facts
- *Artificial Intelligence: A Modern Approach*, Russell and Norvig (4th ed.)
- Chapter 10 – knowledge representation
- Selected review papers on neuro-symbolic AI and agentic AI systems
- Full review-paper citations are listed on the next slide

Main idea: Use the textbooks for the foundations. Use the review papers to see how these ideas are used in current systems.

Review Papers for This Lecture

- DeLong et al. (2025). *Neurosymbolic AI for Reasoning Over Knowledge Graphs: A Survey*.
doi:10.1109/TNNLS.2024.3420218
- Colelough and Regli (2025). *Neuro-Symbolic AI in 2024: A Systematic Review*.
doi:10.48550/arXiv.2501.05435
- Nawaz et al. (2025). *A Review of Neuro-Symbolic AI Integrating Reasoning and Learning for Advanced Cognitive Systems*.
doi:10.1016/j.iswa.2025.200541
- Hakim et al. (2026). *Neuro-Symbolic Agentic AI: Architectures, Integration Patterns, Applications, and Open Challenges*.
doi:10.1016/j.cosrev.2026.100902

Main idea: These review papers give the broader research picture behind the design ideas discussed in this lecture.

Learning Outcomes for Today

By the end of this session, you should be able to:

- explain where a knowledge graph adds value in a modern AI stack
- compare several simple ways to combine graphs with neural models
- explain why tool protocols and multi-agent coordination matter
- describe the role of evidence, guardrails, and traceability in agent systems

Warm-up Question

Question: If an assistant reads only text chunks, what important thing might it miss?

- text chunks may contain the right words
- but they may not show the exact relation
- the answer can become vague or wrong

chunk 1:
Alibaba ...

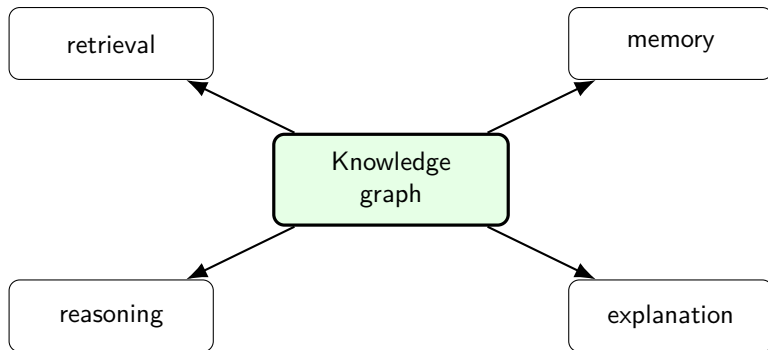
chunk 2:
Hangzhou ...



Main idea: The graph makes the relation explicit. It shows not just the words, but the exact connection between them.

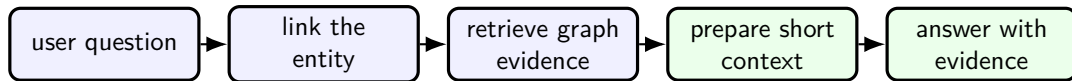
Where Does a Knowledge Graph Help?

Question: In a modern AI system, where can a knowledge graph add value?



Main idea: The graph helps the system find better evidence, remember facts, reason over paths, and explain answers.

Question: How does a knowledge graph improve a retrieval-augmented system?



Main idea: The graph helps retrieval happen before generation, so the answer stays closer to the stored facts.

KGs as Agent Memory

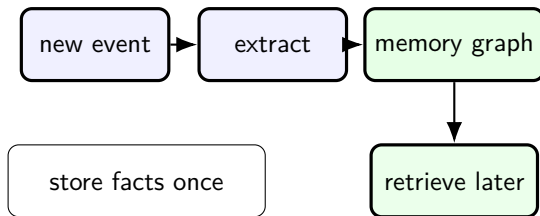
Question: Why is a graph useful when an agent must remember things over many steps?

Without graph memory

The agent keeps facts only in the prompt window.

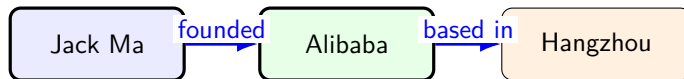
With graph memory

The agent stores entities, links, and updates in a reusable structure.



Main idea: Graph memory helps an agent stay consistent across many tool calls and long tasks.

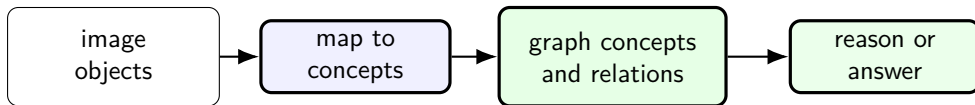
Question: Why are graph paths useful when we want the system to explain an answer?



Evidence path:
Jack Ma → Alibaba → Hangzhou

Main idea: A graph path is a small, readable explanation. It shows where the answer came from.

Question: How can a graph help when the system uses images and text together?



Main idea: The graph gives the model structured concepts and relations, not just raw pixels or raw words.

What Is Neuro-Symbolic AI?

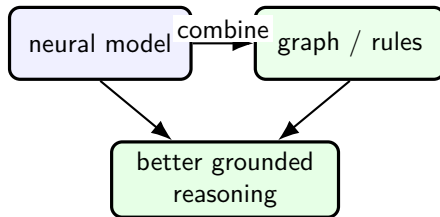
Question: What do people mean when they say neuro-symbolic AI?

Neural part

Good at pattern learning from large data

Symbolic part

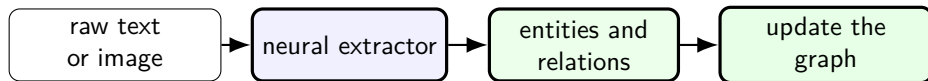
Good at explicit structure, rules, and traceable reasoning



Main idea: Neuro-symbolic AI tries to combine flexible learning with explicit reasoning structure.

Pattern 1: Neural to Symbolic

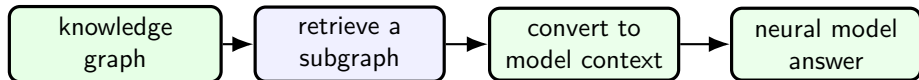
Question: What happens when a neural model writes facts into a graph?



Main idea: Here the neural model reads messy input and turns it into structured graph facts.

Pattern 2: Symbolic to Neural

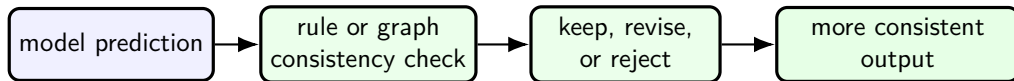
Question: What happens when a graph gives evidence to a neural model?



Main idea: Here the graph guides the neural model by supplying explicit facts before the answer is generated.

Pattern 3: Joint or Constrained

Question: Can we make the neural model and the graph influence each other during learning or checking?



Main idea: This pattern uses graph rules or constraints to catch weak outputs and improve consistency.

What Do Recent Reviews Report?

Question: When people review many neuro-symbolic papers together, what trends do they keep finding?

Often reported benefits

- better explainability
- better multi-hop reasoning
- stronger grounding

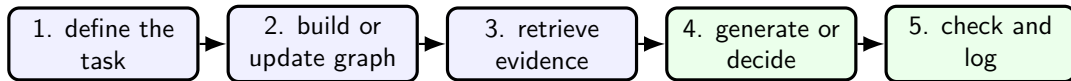
Often reported problems

- incomplete or noisy graphs
- expensive maintenance
- evaluation still inconsistent

Main idea: The idea is promising, but the graph still has to be clean, maintained, and evaluated well.

A Simple Practical Pipeline

Question: If we want to deploy a graph-backed AI system, what is a simple end-to-end workflow?



Main idea: The graph is one stage in a full pipeline, not a magic box by itself.

Operational Challenges

Question: What usually makes a graph-backed AI system difficult to run in practice?

Data problems

entity-linking errors, stale facts, missing source

System problems

retrieval latency, higher cost, harder monitoring

Policy problems

access control, privacy, bias, auditing

Teaching message

strong reasoning still needs careful engineering

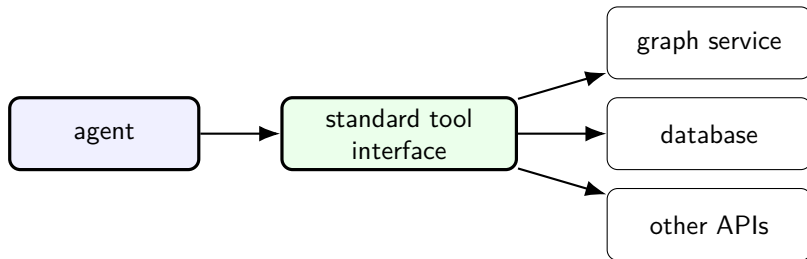
Main idea: Many failures come from bad data, weak controls, or slow retrieval design, not only from the model.

What Is MCP and Why Does It Matter?

Question: What is MCP, and why do modern agent systems need it?

Definition

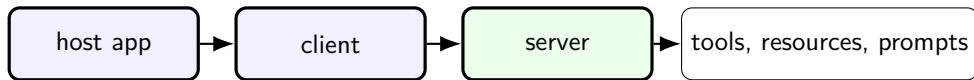
MCP stands for **Model Context Protocol**. It is a standard way for an AI application to connect to external tools, data, and prompts.



Main idea: A standard tool interface makes agent systems easier to connect, swap, and govern.

MCP Core Roles

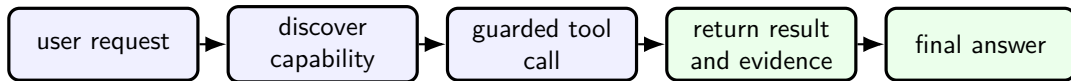
Question: What are the main parts in a simple MCP-style setup?



Main idea: The host runs the assistant, the client connects, and the server exposes usable capabilities.

MCP Request Flow

Question: What happens from a user question to a final grounded answer?



Main idea: The useful idea is simple: discover the right tool, call it safely, then answer with returned evidence.

MCP Security and Governance

Question: What controls do we need before an agent is allowed to use real tools?

Before the call

allowlists, role checks, input checks

After the call

logs, audit trails, policy review

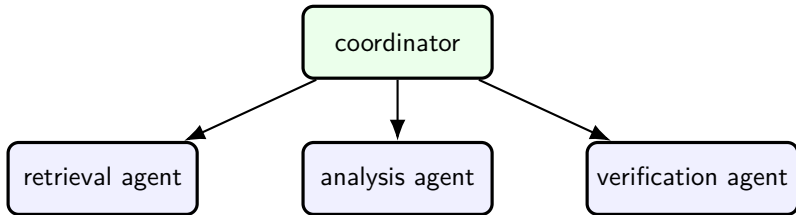
During the call

secrets protection, bounded scope,
timeouts

Main idea: Good tool use is not just about capability. It is also about permission, safety, and traceability.

Why Multi-Agent Systems?

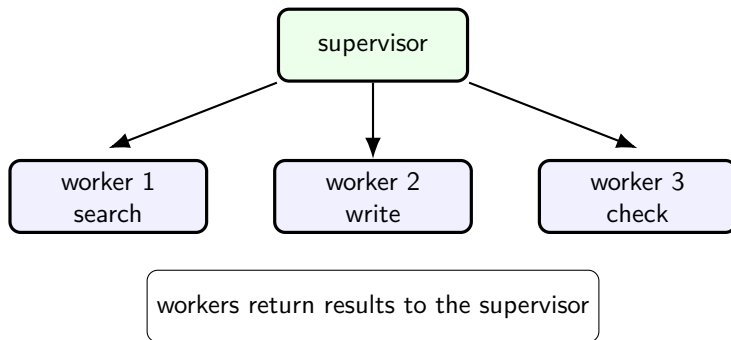
Question: Why use more than one agent instead of one very large prompt?



Main idea: Multiple agents can split work, specialise roles, and check each other.

Pattern: Supervisor and Workers

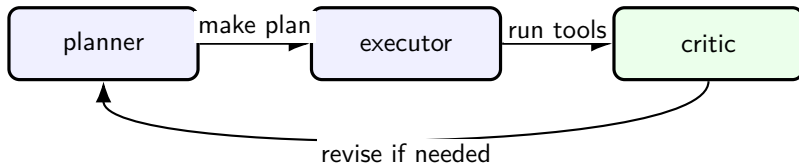
Question: What is the simplest multi-agent pattern to understand first?



Main idea: The supervisor assigns work, workers do one focused job, and the supervisor combines the results.

Pattern: Planner, Executor, Critic

Question: How can we separate planning, doing, and checking?



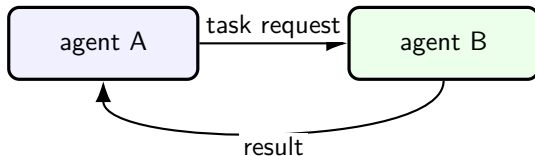
Main idea: This pattern is useful when we want the system to plan first and then check its own work.

What Is Agent-to-Agent (A2A)?

Question: What changes when one agent talks directly to another independent agent?

Simple idea

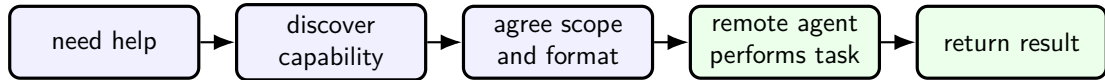
One agent asks another agent for help on a task it cannot or should not do alone.



Main idea: A2A is about coordination across agents, not just tool use inside one agent.

A2A Handoff Sequence

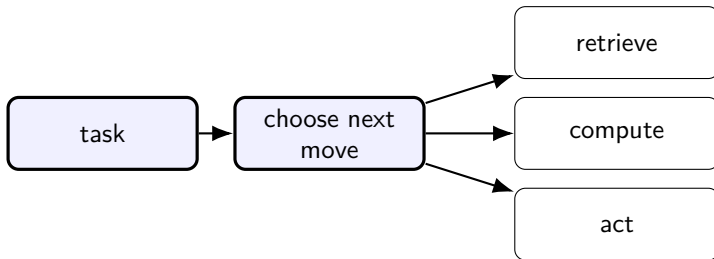
Question: What are the basic steps in a safe agent-to-agent handoff?



Main idea: A good handoff makes the scope clear before work starts, so the result is easier to trust and reuse.

Tool-Choice Reasoning

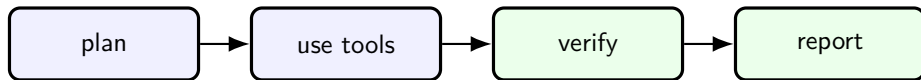
Question: How does a modern assistant decide what to do next?



Main idea: A strong assistant does not call tools randomly. It chooses whether to retrieve, compute, or act.

Coding-Agent Style Runtime Preview

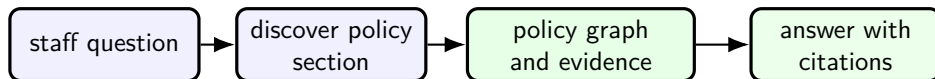
Question: What simple runtime pattern will we study in more detail tomorrow?



Main idea: Tomorrow we will look closely at coding agents, but the basic runtime idea is already here: plan, use tools, verify, report.

Domain Snapshot: Internal Knowledge Copilot

Question: What could a graph-backed enterprise assistant look like in practice?



- retrieve only relevant policy evidence
- keep access control and redaction rules
- return citations with the answer

Main idea: The graph helps the assistant answer with evidence and policy structure, not just fluent text.

Transition to Day 5

Question: What comes after standards, coordination, and graph-backed agent design?

- Today focused on architecture patterns for modern AI systems.
- Tomorrow focuses on coding agents in more detail.
- We will look more closely at trust, tool use, verification, and safe execution.

Main idea: Day 4 showed the system-level picture. Day 5 goes deeper into one important case: coding agents.

Scope Boundary for This Block

Question: What are we deliberately not doing in Block 1?

- We focus on how graphs improve AI behaviour.
- We do not go deep into query-language syntax such as SPARQL or Cypher.
- In this module, query syntax is less important than reasoning flow, grounding, and system design.

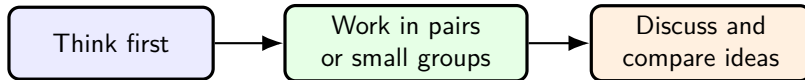
Main idea: For this module, think first about the reasoning system. Treat low-level query syntax as an implementation detail.

Key Takeaways

- Knowledge graphs improve retrieval, memory, reasoning, and explanation.
- Neuro-symbolic systems mix flexible learning with explicit structure.
- MCP, multi-agent patterns, and A2A help organise modern tool-using systems.
- Strong reasoning systems still need evidence, guardrails, and good engineering practice.

Main idea: The big theme is grounding: modern AI becomes more useful when its actions and answers stay connected to explicit evidence and clear system structure.

In-Class Exercises



Use this time to work on today's in-class tasks and prepare one clear answer you can discuss.

Main idea: Use the exercise time to decide where graph structure adds the most value inside a modern AI pipeline.

Further Study

- Read Poole Chapters 16 and 17, and revisit AIMA Chapter 10.
- Practice: Poole Ch.16, Exercises 16.3, 16.4, and 16.5.
- Practice: Poole Ch.17, Exercises 17.4, 17.5, and 17.8.
- AIMA knowledge-representation exercise page: 7, 8, 26, and 28.
- Focus on graph-backed retrieval, structured evidence, and how explicit relations change the reasoning flow.

Full References for This Lecture I

 Russell, S. and Norvig, P. (2021).
Artificial Intelligence: A Modern Approach (4th ed.).
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 Poole, D. L. and Mackworth, A. K. (2023).
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 DeLong, L. N., Mir, R. F. and Fleuriot, J. D. (2024).
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IEEE Transactions on Neural Networks and Learning Systems.
DOI: <https://doi.org/10.1109/TNNLS.2024.3420218>

Full References for This Lecture II



Colelough, E. and Regli, W. C. (2025).

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Computer Science Review, 60, 100902.

DOI: <https://doi.org/10.1016/j.cosrev.2026.100902>

Full References for This Lecture III



Anthropic.

Model Context Protocol specification.

<https://modelcontextprotocol.io/specification/>



Google.

Agent-to-Agent (A2A) protocol documentation.

<https://a2a-protocol.org/dev/>



OpenAI.

OpenAI Agents Python documentation.

<https://openai.github.io/openai-agents-python/>



OpenAI.

Codex product page.

<https://openai.com/codex/>

Full References for This Lecture IV



OpenClaw.

OpenClaw documentation.

<https://getopenclaw.ai/docs>